2. METHODS

2.1. Research Framework

This study applies a structured forecasting and information-system workflow. Daily adjusted closing-price data are acquired and checked for missing or duplicate observations before being divided chronologically into training and testing periods. The complete pipeline is then executed independently for ADRO, PTBA, INDY, and ITMG. Standalone ARIMA was developed as the linear baseline following the Box-Jenkins framework [12] , while standalone LSTM was used as the nonlinear baseline. The hybrid ARIMA-LSTM model combines both approaches by training LSTM on ARIMA residuals, where the predicted residual is added to the ARIMA forecast [13]. The resulting forecasts are evaluated using MAE, RMSE, MAPE, and R^2, after which the company-specific hybrid ARIMA-LSTM models generate exploratory monthly projections for 2025–2045 that are presented through a web-based interactive dashboard [14].

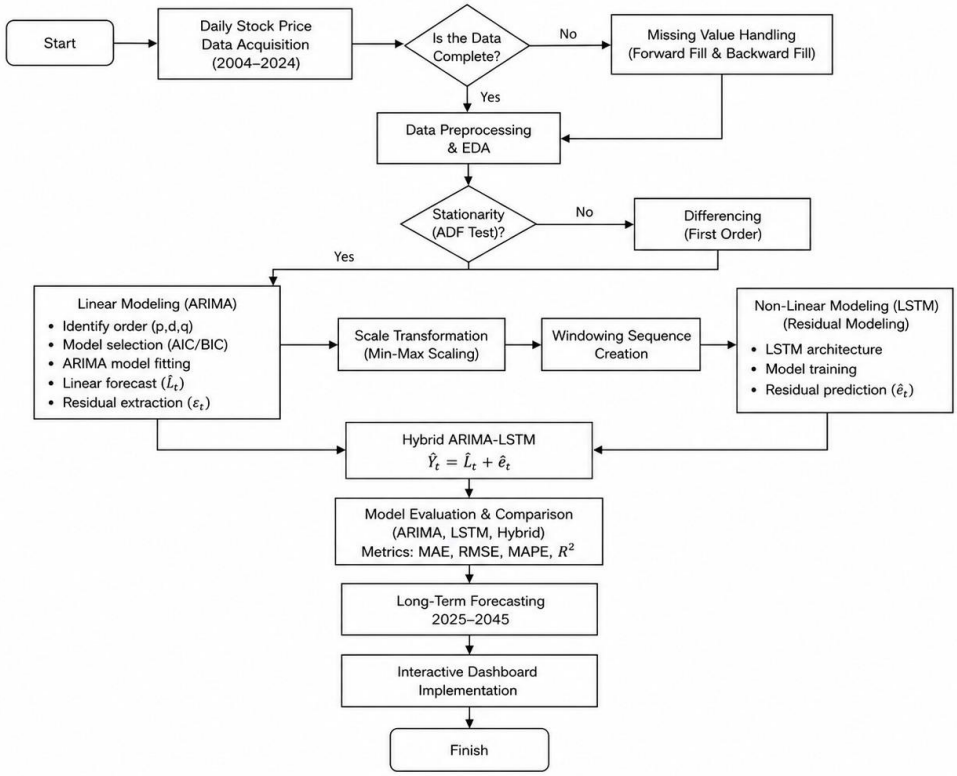


Figure 1. Research workflow of the hybrid ARIMA-LSTM model for coal-stock price forecasting.

2.2. Research Data and Dataset Split

The data used in this study consist of daily adjusted closing prices obtained from Yahoo Finance historical market data using the Python library yfinance. The dataset includes four Indonesian coal-sector stocks listed on the Indonesia Stock Exchange (IDX), namely ADRO (ADRO.JK), PTBA (PTBA.JK), INDY (INDY.JK), and ITMG (ITMG.JK). The dataset consists of daily trading observations (business days) from January 2004 to December 2024. Yahoo Finance provides several attributes, including Open, High, Low, Close, Adjusted Close, and Volume. Among these attributes, only the Adjusted Closing Price (Adj Close) was used as the forecasting input. The use of adjusted closing prices provides a more consistent historical series because it accounts for corporate actions such as stock splits and dividend adjustments. This variable is commonly used in univariate financial time-series forecasting because it allows models to focus on temporal patterns from historical price movements. Furthermore, using a single input variable ensures that ARIMA, LSTM, and hybrid ARIMA-LSTM models are evaluated under the same input conditions.

The collected dataset contains daily stock-price observations with the variables Date, ADRO, PTBA, INDY, and ITMG, while daily return variables were calculated to analyze price changes. The exact observation periods used in this study were July 16, 2008–December 30, 2024 for ADRO; January 2, 2004–December 30, 2024 for PTBA; June 11, 2008–December 30, 2024 for INDY; and December 18, 2007–December 30, 2024 for ITMG. The different starting dates occur because each stock has different historical data availability in Yahoo Finance.

However, this approach does not include external factors such as coal commodity prices, exchange rates, macroeconomic indicators, trading volume, or company fundamentals, which may also influence stock-price fluctuations [15]. This univariate approach was selected to evaluate the intrinsic predictive capability of historical price patterns before incorporating additional external variables in future multivariate forecasting frameworks.

The four companies were not combined into a single multivariate model; instead, each ticker was processed, trained, and evaluated independently using its own historical series. The dataset was divided using a chronological split with a calendar cutoff of December 31, 2022. Therefore, observations until December 30, 2022 were used as training data, while observations from January 2, 2023 to December 30, 2024 were used as testing data.

Furthermore, 15% of the training data were separated as validation data during LSTM training to monitor model performance and implement early stopping. This splitting strategy prevents future information leakage and ensures that evaluation is performed using unseen observations.

Table 1. Research dataset

Company	Ticker	Full observation period	Training period	Testing period
ADRO	ADRO.JK	16 Jul. 2008-30 Dec. 2024	16 Jul. 2008-30 Dec. 2022	2 Jan. 2023-30 Dec. 2024
PTBA	PTBA.JK	2 Jan. 2004-30 Dec. 2024	2 Jan. 2004-30 Dec. 2022	2 Jan. 2023-30 Dec. 2024
INDY	INDY.JK	11 Jun. 2008-30 Dec. 2024	11 Jun. 2008-30 Dec. 2022	2 Jan. 2023-30 Dec. 2024
ITMG	ITMG.JK	18 Dec. 2007-30 Dec. 2024	18 Dec. 2007-30 Dec. 2022	2 Jan. 2023-30 Dec. 2024

2.3. Data Preprocessing

Data preprocessing was performed independently for each stock to prepare the historical price series before ARIMA, LSTM, and hybrid ARIMA-LSTM modeling. The preprocessing procedure consisted of data quality checking, return calculation, stationarity testing, differencing, normalization, and sequence preparation.

First, the dataset was examined for missing values, duplicate records, and unusual values to ensure data consistency [16]. The observations were arranged chronologically based on trading dates [17]. Missing values, if present, were handled using forward fill and backward fill methods to maintain continuity in the time-series data. Daily return values were calculated from consecutive adjusted closing prices to analyze relative price changes; however, the forecasting models used adjusted closing prices as the main input variable.

For ARIMA modeling, stationarity testing was conducted using the Augmented Dickey-Fuller (ADF) test and confirmed using the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test. Since the stock-price series at level were identified as non-stationary, first-order differencing was applied to obtain a stationary series. The differencing process is defined as:

$$\Delta P_t = P_t - P_{\{t-1\}} \quad (1)$$

where P_t represents the closing price at time t and ΔP_t represents the differenced price series. Based on the stationarity testing results, the integration order used in ARIMA modeling was set to $d = 1$.

For the hybrid ARIMA-LSTM model, the residual values generated from the ARIMA forecasting process were used as input for the LSTM model [18]. The residual sequence was transformed using Min-Max scaling with a range of $(-1,1)$ to normalize the values into a consistent range. The scaling process was performed using training data and then applied to validation and testing data to prevent information leakage [19]. All preprocessing stages were conducted separately for ADRO, PTBA, INDY, and ITMG to preserve company-specific price characteristics and volatility patterns. The normalized residual data were converted into supervised learning sequences using a sliding-window approach [20]. A window size of 30 previous observations was used as input to predict the following residual value.

2.4. ARIMA Model Development

ARIMA was used to capture the linear component of each stock-price series. The ARIMA model is represented as $ARIMA(p,d,q)$, where p represents the autoregressive order, d represents the differencing order, and q represents the moving-average order. The general formulation of ARIMA is expressed as:

$$\phi^{d}_{p(B)(1-B)} P_t = c + \theta_{q(B)\varepsilon_t} \quad (2)$$

formulation is expressed in Equation (2), where B denotes the backshift operator, d is the differencing order, $\phi_{p(B)}$ represents the autoregressive polynomial, $\theta_{q(B)}$ represents the moving-average polynomial, c is a constant, and ϵ_t is the random error.

The ARIMA parameters were determined separately for each company because each stock has different historical price characteristics. The differencing order (d) was determined based on the stationarity testing results described in the preprocessing stage. Since all stock-price series were identified as non-stationary at the original level, first-order differencing was applied, resulting in an integration order of $d = 1$.

The autoregressive order (p) and moving-average order (q) were selected using a candidate model evaluation procedure based on grid search and Akaike Information Criterion (AIC) [21]. Several combinations p and q were fitted using the training dataset, with both parameters evaluated within the range of 0 to 4. The model with the lowest AIC value was selected as the optimal ARIMA configuration because it provides a balance between model fit and complexity. The parameter selection process was conducted independently for ADRO, PTBA, INDY, and ITMG.

The selected configurations are summarized in Table 2.

Table 2. Selected ARIMA configurations for each company

Company	Ticker	Selected ARIMA order	Model role
ADRO	ADRO.JK	ARIMA(3,1,3)	Linear price forecast
PTBA	PTBA.JK	ARIMA(1,1,3)	Linear price forecast
INDY	INDY.JK	ARIMA(1,1,3)	Linear price forecast
ITMG	ITMG.JK	ARIMA(3,1,3)	Linear price forecast

The ARIMA-only output represents the linear forecast generated from the selected company-specific ARIMA model. The forecast values and residuals generated from the ARIMA were subsequently used in the hybrid ARIMA-LSTM stage, where LSTM was applied to learn nonlinear patterns from the residual component [22].

2.5. LSTM Model Development

Long Short-Term Memory (LSTM) was used to model nonlinear temporal patterns in stock-price movements. Unlike ARIMA, which captures linear dependencies in time-series data, LSTM is designed to learn sequential patterns through its memory mechanism [23]. In this study, LSTM was developed as a standalone forecasting model and was also used as the nonlinear component in the hybrid ARIMA-LSTM framework.

The standalone LSTM used adjusted closing-price sequences with a 30-day sliding window to predict the next value [24]. The architecture consisted of two LSTM layers followed by dropout and dense layers, while model training used the Adam optimizer, MSE loss, validation data, and early stopping. The LSTM hyperparameters were fixed and applied consistently across all four stocks to maintain model comparability. The validation subset was used to monitor training through early stopping and adaptive learning-rate reduction rather than for hyperparameter search. The forecasting framework was implemented in Python using yfinance for data acquisition, statistical and machine-learning libraries for preprocessing and modelling, and TensorFlow/Keras for LSTM training, with a fixed random seed of 2024 applied to support reproducibility.

The LSTM architecture and training configuration are summarized in Table 3.

Table 3. LSTM architecture and training configuration

Component	Configuration
Input window	30 daily observations
First LSTM layer	64 units; return sequences enabled
Dropout after first LSTM	0.20
Second LSTM layer	64 units
Dropout after second LSTM	0.20
Dense layer	32 units; ReLU activation
Output layer	1 unit; linear output
Total trainable parameters	52,033
Optimizer and learning rate	Adam; 0.001
Loss function	Mean Squared Error
Batch size and maximum epochs	32; 100 epochs
Validation and early stopping	15% validation split; patience = 15

2.6. LSTM and Hybrid ARIMA-LSTM Development

The hybrid ARIMA-LSTM model combines the linear forecasting capability of ARIMA with the nonlinear sequence learning capability of LSTM [25].

The residual component was calculated as:

$$e_t = P_t - P_t^{ARIMA} \tag{3}$$

where P_t represents the observed stock price at time t, and P_t^{ARIMA} represents the ARIMA forecast. The residuals were normalized and transformed into 30-day input sequences, while LSTM estimated the nonlinear residual correction [26].

The final hybrid forecast was obtained as:

$$P_t^{Hybrid} = P_t^{ARIMA} + e_t^{LSTM} \tag{4}$$

where P_t^{Hybrid} represents the final hybrid prediction and e_t^{LSTM} represents the nonlinear correction estimated by LSTM. The model was developed independently for ADRO, PTBA, INDY, and ITMG using the same chronological data split.

Standalone ARIMA and LSTM models were also evaluated as baseline comparisons using the same dataset division and evaluation procedure to assess whether hybridization improved forecasting performance.

2.7. Model Evaluation and Performance Measurement

Model performance was evaluated on the testing period by comparing the forecasting results of ARIMA, LSTM, and hybrid ARIMA-LSTM models using three error-based evaluation metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). These metrics were selected because they provide complementary measures of forecasting accuracy across different error characteristics.

MAE measures the average absolute difference between the predicted and observed prices, while RMSE assigns a higher penalty to larger forecasting errors. MAPE expresses the prediction error in percentage terms, allowing proportional comparison across stocks with different price scales. Lower values of MAE, RMSE, and MAPE indicate better forecasting performance.

The evaluation metrics are defined as follows:

$$MAE = \left(\frac{1}{n}\right)\sum_{t=1}^{n}|y_t - y_t| \tag{5}$$

$$RMSE = \sqrt{\left(\frac{1}{n}\right)\sum_{t=1}^{n}(y_t - y_t)^2} \tag{6}$$

$$MAPE = \left(\frac{100\%}{n}\right)\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right| \tag{7}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{8}$$

where y_t represents the observed stock price, $\hat{y}_t$represents the predicted value, and n represents the number of observations in the testing period.

The evaluation was conducted separately for ADRO, PTBA, INDY, and ITMG. The forecasting accuracy of each model was then compared using a performance comparison table containing the MAE, RMSE, and MAPE values for ARIMA, LSTM, and hybrid ARIMA-LSTM models. The primary evaluation used a chronological holdout strategy because financial time-series forecasting requires preserving temporal order between training and testing data. The selected approach evaluates model performance on unseen observations while preventing future information leakage [27]. To assess robustness beyond the final holdout period, an expanding-window validation strategy was additionally applied using three pre-test evaluation windows. The models were trained using observations available up to the

end of 2019, 2020, and 2021 and were subsequently evaluated on 2020, 2021, and 2022, respectively. All preprocessing and model estimation procedures were repeated independently within each fold to prevent future information leakage, while the January 2023–December 2024 period remained untouched for the final out-of-sample evaluation.

For the 2025–2045 exploratory projections, the ARIMA component was re-estimated using the full available price series through December 2024. Forecasts and 95% confidence intervals were generated using the SARIMAX get_forecast() and conf_int(alpha=0.05) procedures, respectively. These confidence bounds represent uncertainty associated with the ARIMA component and are used as an uncertainty envelope for the hybrid projections; they do not incorporate additional uncertainty from the LSTM residual correction.